On the directions occurring in lattice-line coverings of the integer plane[1]

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Abstract

We consider families of lines that cover every point of the integer lattice $\mathbb{Z}^2$, subject to the constraint that no two lines of different direction in the family meet at a lattice point. Restricting to *lattice lines* (lines containing at least two, hence infinitely many, lattice points — equivalently, of rational direction), we show that the set of directions occurring in such a covering can be made dense in the space of line directions. The construction is a recursive splitting of $\mathbb{Z}^2$ into nested rank-2 sublattice cosets, each handed off to a freshly chosen direction; the key technical point is a steering lemma showing that at every stage of the recursion a new direction arbitrarily close to any prescribed target can still be realized, via an elementary sieve bound.

1. Introduction

Consider the integer lattice $\mathbb{Z}^2 \subset \mathbb{R}^2$. We wish to cover every point of $\mathbb{Z}^2$ by a family $\mathcal{F}$ of lines, subject to: any two lines of $\mathcal{F}$ may cross, but never at a point of $\mathbb{Z}^2$. Which sets of line-directions can occur across such a family?

Remark 1 (The lattice-line hypothesis). As posed, with no further restriction on the lines allowed, the question is trivial: for each $z \in \mathbb{Z}^2$ take the line L_z through z of irrational slope α_z. Then

$$L_z \cap \mathbb{Z}^2 = \{z\} \tag{1}$$

(a second lattice point on L_z would force $\alpha_z \in \mathbb{Q}$), so distinct such lines never share a lattice point, and choosing the α_z to range over a dense (indeed, all) set of irrational directions already realizes an uncountable dense set of directions. Everything of interest in this note therefore concerns the restriction to *lattice lines*: lines containing at least two (hence, by the same argument, infinitely many) points of $\mathbb{Z}^2$, equivalently lines of rational direction. This restriction is in force throughout.

We now formalize the objects of study as explicit definitions, rather than leaving them as prose, so that every later lemma statement can refer back to them unambiguously.

Definition 2 (Primitive direction). A *primitive direction* is a pair $d = (p, q) \in \mathbb{Z}^2$ with $\gcd(p, q) = 1$. Two primitive directions d, d' are *equal as directions* if $d' = \pm d$. Write

$$\varphi_d(x, y) = qx - py, \tag{2}$$

a surjective homomorphism $\mathbb{Z}^2 \to \mathbb{Z}$ depending only on d up to sign of the whole map (in particular $\ker \varphi_d = \ker \varphi_{-d}$). The space of directions is $\mathbb{RP}^1 = S^1/\{\pm 1\}$, parametrized by $\theta \in [0, \pi)$; we metrize it by

$$d(\theta, \theta') = \min(|\theta - \theta'|, \pi - |\theta - \theta'|), \tag{3}$$

so "within ε of θ" always means within ε in this metric. Since $[0, \pi) \to \mathbb{RP}^1$ is a continuous surjection, a set dense in $[0, \pi)$ is automatically dense in $\mathbb{RP}^1$.

Definition 3 (Lattice lines). For a primitive direction $d = (p, q)$ and $c \in \mathbb{Z}$, the *lattice line* $\ell_{d,c}$ is the level set

$$\ell_{d,c} = \varphi_d^{-1}(c) = \{(x, y) \in \mathbb{Z}^2 : qx - py = c\}. \tag{4}$$

Every line in $\mathbb{R}^2$ containing at least two points of $\mathbb{Z}^2$ equals $\ell_{d,c}$ for a unique primitive direction d (up to sign) and a unique $c \in \mathbb{Z}$; we call such a line a lattice line of direction d.

Definition 4 (Covering family, valid family). A *covering family* is a set $\mathcal{F}$ of lattice lines with $\bigcup \mathcal{F} \supseteq \mathbb{Z}^2$. A covering family $\mathcal{F}$ is *valid* if no two lines of $\mathcal{F}$ of different direction share a point of $\mathbb{Z}^2$; equivalently, for every pair of distinct primitive directions d, d' occurring in $\mathcal{F}$, and every $\ell_{d,c}, \ell_{d',c'} \in \mathcal{F}$, we have $\ell_{d,c} \cap \ell_{d',c'} \cap \mathbb{Z}^2 = \varnothing$.

One easily disposes of the variant where lines are required to be pairwise disjoint everywhere (not merely off the lattice): two distinct lines in $\mathbb{R}^2$ are either parallel or meet at exactly one point, so pairwise disjointness forces every line in the family to share a single direction, and any one direction already suffices (Example 5 below). All the content is in the "crossings allowed off the lattice" variant of Definition 4, which is our subject here.

[1]This is a working-notes variant of the formal draft `lattice_line_covers_formal.tex` (itself a more formal, more detailed sibling of the canonical short draft `lattice_line_covers.tex`; both are unchanged by this version). The only content added here is a red *Pedantic example* environment immediately after each lemma's statement, giving fully numeric input values for which the conclusion holds, and fully numeric input values violating one of the lemma's hypotheses, to make concrete why that hypothesis cannot be dropped. These examples are for Jan Snellman's own benefit while reading the proofs and are deliberately *not* intended to appear in any final version of this article. Separately, the statement, proof, and Lean-status note of the Rigidity lemma (Lemma 7) have been reworded to index the two directions by $1, 2$ rather than i, j: the original i, j notation suggested, misleadingly, that the two directions range over some indexed family, when the lemma is really about an arbitrary fixed *pair* of directions.

Example 5. For a primitive direction $d = (p, q)$, the lines $\ell_{d,c}$, $c \in \mathbb{Z}$, partition $\mathbb{Z}^2$ (they are the fibers of the surjective homomorphism φ_d):

$$\mathbb{Z}^2 = \bigsqcup_{c \in \mathbb{Z}} \ell_{d,c}. \tag{5}$$

So a single direction always suffices to cover $\mathbb{Z}^2$ by pairwise-disjoint (in particular lattice-point-disjoint) lattice lines.

Our main result:

Theorem 6. *There is a valid covering family $\mathcal{F}$ (Definition 4) whose set of realized directions is dense in $\mathbb{RP}^1$.*

Lean formalization status: not yet formalized. This theorem is the final assembly of Lemmas 10, 12, and 13, via an infinite recursive/diagonal construction requiring well-founded recursion with an existential (choice-dependent) witness at each step. Formalizing this construction, and Lemma 13 itself, are the two pieces of work explicitly flagged as the hardest remaining parts of this formalization project before this session's work on Lemma 13 even began.

2. Two lemmas on lattice lines

Lemma 7 (Rigidity). *Let $d_1 = (p_1, q_1)$, $d_2 = (p_2, q_2)$ be* two *distinct primitive directions — not members of an indexed family; there is no set that* $1, 2$ *range over here, they are simply labels for "the first direction" and "the second direction" — and set*

$$\Delta = p_1 q_2 - p_2 q_1 \neq 0. \tag{6}$$

The lines ℓ_{d_1,c_1} and ℓ_{d_2,c_2} meet at a lattice point if and only if

$$\Delta \mid (p_1 c_2 - p_2 c_1) \quad \text{and} \quad \Delta \mid (q_1 c_2 - q_2 c_1). \tag{7}$$

In particular, if $|\Delta| = 1$, condition (7) holds vacuously for every *choice of c_1, c_2, so d_1, d_2 can never both occur in a valid family (Definition 4).*

Pedantic example (working notes only — not for the final article). (i) A unimodular pair ($|\Delta| = 1$): forced for every c_1, c_2. Take $d_1 = (1, 0)$, $d_2 = (0, 1)$, so $\Delta = 1 \cdot 1 - 0 \cdot 0 = 1$. For *any* c_1, c_2, say $c_1 = 17$, $c_2 = -4$: the system reads $-y = 17$, $x = -4$, giving the lattice point $(x, y) = (-4, -17)$ regardless of what c_1, c_2 were — this is why $(1, 0)$ and $(0, 1)$ (or any Farey-neighbor pair) can never both appear in a valid family, whatever levels are chosen. **(ii) A non-unimodular pair, divisibility holding (conclusion: they meet).** Take $d_1 = (1, 0)$, $d_2 = (1, 2)$, so $\Delta = 1 \cdot 2 - 1 \cdot 0 = 2$. With $c_1 = 0$, $c_2 = 4$: $p_1 c_2 - p_2 c_1 = 4$ and $q_1 c_2 - q_2 c_1 = 0$, both divisible by $\Delta = 2$, so the lemma predicts a lattice crossing; indeed the system $-y = 0$, $2x - y = 4$ gives $(x, y) = (2, 0) \in \mathbb{Z}^2$. **(iii) Same pair, divisibility failing (conclusion: they do not meet).** Same d_1, d_2, but $c_1 = 0$, $c_2 = 3$: $p_1 c_2 - p_2 c_1 = 3$ is *not* divisible by $\Delta = 2$, so no lattice crossing should occur; indeed the system $-y = 0$, $2x - y = 3$ gives $x = 3/2 \notin \mathbb{Z}$. **(iv) Why $\Delta \neq 0$ (i.e. $d_1 \neq d_2$) cannot be dropped.** Take $d_1 = d_2 = (1, 0)$, so $\Delta = 0$; equations (9)–(10) below would read $0 \cdot x = p_1 c_2 - p_2 c_1$, undefined as a formula for x unless the right side is also 0. Concretely, with $c_1 = 0 \neq c_2 = 1$ the two lines $-y = 0$ and $-y = 1$ are parallel and distinct, hence never meet at all (not even off the lattice); with $c_1 = c_2 = 0$ the two lines coincide, sharing *every* lattice point on the line, not a single isolated crossing — either way the lemma's "if and only if a single congruence condition holds" shape breaks down, which is exactly why distinctness of the two directions is a standing hypothesis.

Proof. A lattice point (x, y) lies on both lines exactly when the linear system

$$\begin{aligned} q_1 x - p_1 y &= c_1, \\ q_2 x - p_2 y &= c_2 \end{aligned} \tag{8}$$

holds. Eliminating y (multiply the first equation by p_2, the second by p_1, and subtract) gives

$$\Delta x = p_1 c_2 - p_2 c_1, \tag{9}$$

and eliminating x symmetrically gives

$$\Delta y = q_1 c_2 - q_2 c_1. \tag{10}$$

Since $\Delta \neq 0$, equations (9)–(10) determine $x, y \in \mathbb{Q}$ uniquely (this is exactly Cramer's rule applied to (8)), and $(x, y) \in \mathbb{Z}^2$ if and only if Δ divides both right-hand sides, which is precisely (7). Conversely, if (7) holds, the quotients in (9)–(10) are integers and one checks directly, by substituting back into (8), that they solve the system: this is a direct algebraic identity, not merely a necessary condition. If $|\Delta| = 1$, both divisibility conditions in (7) hold automatically for every $c_1, c_2 \in \mathbb{Z}$. □

Lean formalization status: ***formally verified.*** *Theorem `LatticeLineCovers.rigidity` in `lean/LatticeLineCoversLean/Basic.lean`. (The Lean source itself names its six integer parameters `p_i q_i p_j q_j c_i c_j` — purely as identifier characters, with no indexed family behind them either; this note's renaming to $1, 2$ is a presentational choice for this article, not a claim that the Lean statement disagrees with it.) Both directions are proved by `linear_combination` (an automated tactic verifying a claimed polynomial-ring identity), matching exactly the elimination computation in the proof above; the reverse direction needs one additional cancellation step (`mul_left_cancel₀`) since $\Delta \neq 0$ must be used to recover x, y from (9)–(10). Independently re-verified to depend only on Lean's three standard axioms (`propext`, `Classical.choice`, `Quot.sound`), i.e. on no unproved `sorry`. Notably, the Lean statement requires* no *primitivity/coprimality hypothesis on d_1, d_2 at all — confirmed by deliberately adding $\gcd(p_1, q_1) = \gcd(p_2, q_2) = 1$ as hypotheses and observing, via Lean's own unused-variable linter, that the existing proof never refers to them. Primitivity is needed only for Definition 2's bijection between directions and pairs (p, q), not for this lemma's algebraic content.*

Lemma 8 (Coset). *Let p, q be coprime integers, $x_0, y_0 \in \mathbb{Z}$, and set*

$$c_0 = qx_0 - py_0. \tag{11}$$

Then

$$\bigcup_{k\in\mathbb{Z}} \ell_{(p,q),c_0+pqk} = \{(x,y)\in\mathbb{Z}^2 : x \equiv x_0 \pmod{|p|},\ y \equiv y_0 \pmod{|q|}\}. \tag{12}$$

Pedantic example (working notes only — not for the final article). (i) Conclusion holds. Take $p = 2$, $q = 3$ (coprime), $x_0 = y_0 = 1$, so $c_0 = 3 \cdot 1 - 2 \cdot 1 = 1$ and $pq = 6$. The claimed set is $\{x \text{ odd},\ y \equiv 1 \pmod 3\}$. Level $k = 0$ ($c = 1$): the point $(1,1)$ satisfies $3 \cdot 1 - 2 \cdot 1 = 1$, and indeed 1 is odd, $1 \equiv 1 \pmod 3$. Level $k = 1$ ($c = 7$): the point $(3,1)$ satisfies $3 \cdot 3 - 2 \cdot 1 = 7$, and again 3 is odd, $1 \equiv 1 \pmod 3$. Both lie in C as claimed, at the levels the formula predicts. **(ii) Why $\gcd(p,q) = 1$ cannot be dropped.** Take the non-coprime pair $p = 2, q = 4$ ($\gcd = 2$), $x_0 = y_0 = 0$, so $c_0 = 0$ and $pq = 8$. The point $(x,y) = (1,2)$ satisfies $qx - py = 4 \cdot 1 - 2 \cdot 2 = 0 = c_0 + 8 \cdot 0$, i.e. it lies on the level-0 line, exactly as a point of the claimed union should — but $x = 1$ is *odd*, so it violates $x \equiv x_0 = 0 \pmod p) = 2$, the very congruence the lemma claims the union equals. So with p, q not coprime the conclusion is simply false, not just unproved: the union of levels is a strictly larger set than the claimed congruence class, because $\varphi_{(2,4)}$ is not surjective onto $\mathbb{Z}$ in the way the coprime case relies on ($\gcd(p,q)$ divides every value of $qx - py$, so the "level $c_0 + pqk$" spacing no longer lines up with the individual mod-p, mod-q congruences).

Proof. Write C for the right-hand side of (12). Since $\gcd(p,q) = 1$ implies in particular $p, q \neq 0$, congruence modulo p (resp. q) is well defined, and C admits the parametrization

$$C = \{(x_0 + pu,\ y_0 + qv) : u, v \in \mathbb{Z}\}. \tag{13}$$

For a point of the form (13),

$$\varphi_{(p,q)}(x_0 + pu,\ y_0 + qv) = q(x_0 + pu) - p(y_0 + qv) = c_0 + pq(u - v), \tag{14}$$

using (11) and cancelling the pqu terms. By (4), this point lies on $\ell_{(p,q),c_0+pqk}$ if and only if $u - v = k$. As u ranges over $\mathbb{Z}$ (with $v = u - k$ then determined), every point of C with that particular value of $u - v$ is visited exactly once, and (14) shows every point of $\ell_{(p,q),c_0+pqk} \cap C$ has $u - v = k$: hence

$$\ell_{(p,q),c_0+pqk} \cap C = \{(x_0 + pu,\ y_0 + q(u - k)) : u \in \mathbb{Z}\}, \tag{15}$$

and moreover every point of $\ell_{(p,q),c_0+pqk}$ manifestly lies in C (take u, v with $u - v = k$ so that (14) matches $c_0 + pqk$; such (x,y) automatically satisfies $x \equiv x_0 \pmod p$, $y \equiv y_0 \pmod q$). So $\ell_{(p,q),c_0+pqk} \subseteq C$ for every k, with equality of the union to all of C once k ranges over $\mathbb{Z}$, since every pair $(u,v) \in \mathbb{Z}^2$ has *some* value of $u - v$. □

Lean formalization status: ***formally verified.*** *Theorem* `LatticeLineCovers.coset`*, same file. Stated using Mathlib's* `IsCoprime` *predicate (a Bézout-identity characterization,* $\exists a, b : ap + bq = 1$*, rather than a computed* $\gcd$ *value) for the hypothesis* $\gcd(p,q) = 1$*, and Mathlib's* `Int.ModEq` *(notation* `a ≡ b [ZMOD n]`*) for the congruences — the latter is convenient here because* `Int.ModEq` *does not depend on the sign of the modulus, so* `[ZMOD p]` *already means "modulo* $|p|$*" with no extra bookkeeping. The forward direction of the proof is a coprimality-transfer argument (from* $p \mid q(x_0 - x)$ *and* $\gcd(p,q) = 1$*, conclude* $p \mid (x_0 - x)$*, via* `IsCoprime.dvd_of_dvd_mul_left`*); the reverse direction extracts explicit witnesses from the two congruences and computes* k *directly, matching (14). Independently re-verified axiom-clean. Proved by a subagent instance of Claude running on the Opus model, briefed with a written proof plan derived from the proof above; compiled successfully on the first attempt.*

The rigidity lemma shows that a family realizing many directions cannot simply throw them together: directions related by a determinant of ± 1 ("unimodular pairs", e.g. any pair of Farey neighbors read as slopes) are permanently incompatible. The coset lemma is the tool that lets us build a family avoiding all bad pairs at once, applied recursively below.

3. The construction

Definition 9 (Rank-2 coset). For coprime positive integers n_1, n_2 and $x_0, y_0 \in \mathbb{Z}$, define

$$R(n_1, n_2, x_0, y_0) = \{(x,y) \in \mathbb{Z}^2 : x \equiv x_0 \pmod{n_1},\ y \equiv y_0 \pmod{n_2}\}. \tag{16}$$

Since $n_1, n_2 > 0$, every $(x,y) \in R(n_1, n_2, x_0, y_0)$ is of the form $x = x_0 + n_1 u$, $y = y_0 + n_2 w$ for a *unique* pair $(u,w) \in \mathbb{Z}^2$; call (u,w) the *internal coordinates* of the point.

We now state the two halves of the original Splitting lemma as two separate lemmas, matching a corresponding split made when formalizing this argument in Lean (see the status notes below): the first identifies each finer sub-coset with a plain coset of a new direction, and the second records that these sub-cosets genuinely partition R.

Lemma 10 (Splitting: coset identity). *Let* $R = R(n_1, n_2, x_0, y_0)$*, let* s, t *be nonzero integers with*

$$\gcd(s, n_2) = \gcd(t, n_1) = \gcd(s, t) = 1, \tag{17}$$

and put

$$P = n_1 s, \quad Q = n_2 t. \tag{18}$$

For $u_0, w_0 \in \mathbb{Z}$*, let*

$$R_{u_0,w_0} = \{(x,y) \in R : x = x_0 + n_1 u,\ y = y_0 + n_2 w \text{ for some } u \equiv u_0 \pmod{|s|},\ w \equiv w_0 \pmod{|t|}\} \tag{19}$$

(a sub-coset of R, picked out via the internal coordinates of Definition 9). Then

$$\bigcup_{k\in\mathbb{Z}} \ell_{(P,Q),c_0+PQk} = R_{u_0,w_0}, \quad c_0 = Q(x_0 + n_1 u_0) - P(y_0 + n_2 w_0). \tag{20}$$

In particular (P,Q) is itself a primitive direction: $\gcd(P,Q) = 1$.

Pedantic example (working notes only — not for the final article). (i) Conclusion holds. Take $n_1 = 2, n_2 = 3, x_0 = y_0 = 1$ (so $R = \{x \text{ odd},\ y \equiv 1 \,(\mathrm{mod}\, 3)\}$), $s = 2, t = 3$; one checks $\gcd(s,n_2) = \gcd(2,3) = 1$, $\gcd(t,n_1) = \gcd(3,2) = 1$, $\gcd(s,t) = \gcd(2,3) = 1$, so (17) holds, and $P = 4, Q = 9$ with $\gcd(4,9) = 1$ as claimed. (Full numeric trace with three explicit points: Example 11 below.) **(ii) Why $\gcd(s,n_2) = 1$ cannot be dropped.** Same $n_1 = 2, n_2 = 3$, but take $s = 3, t = 1$: then $\gcd(t,n_1) = \gcd(1,2) = 1$ and $\gcd(s,t) = \gcd(3,1) = 1$ both still hold, but $\gcd(s,n_2) = \gcd(3,3) = 3 \neq 1$, a violation of just the *first* condition of (17) in isolation. Then $P = n_1 s = 6, Q = n_2 t = 3$, and $\gcd(P,Q) = \gcd(6,3) = 3 \neq 1$: (P,Q) is *not* a primitive direction — exactly because the proof's "primitivity of (P,Q)" case analysis needs $\gcd(s,n_2) = 1$ to rule out a prime (here 3) dividing both s and n_2 simultaneously, and hence dividing both $P = n_1 s$ and $Q = n_2 t$ at once.

Proof. Primitivity of (P,Q). A prime dividing both $P = n_1 s$ and $Q = n_2 t$ would have to divide one of n_1, n_2 (excluded, as $\gcd(n_1,n_2) = 1$), or n_1 and t (excluded by $\gcd(t,n_1) = 1$ in (17)), or s and n_2 (excluded by $\gcd(s,n_2) = 1$), or s and t (excluded by $\gcd(s,t) = 1$). These four cases are exhaustive, so no such prime exists.

The identity (20). For $(x,y) = (x_0 + n_1 u,\ y_0 + n_2 w)$,

$$Qx - Py = Q(x_0 + n_1 u) - P(y_0 + n_2 w) = (Qx_0 - Py_0) + n_1 n_2 (tu - sw), \tag{21}$$

using $Q = n_2 t$, $P = n_1 s$. By Lemma 8 applied to the coprime pair (s,t) in the *internal* coordinates (u,w),

$$\bigcup_{k\in\mathbb{Z}} \{(u,w) : tu - sw = c_0^{\text{loc}} + stk\} = \{u \equiv u_0 \pmod{|s|},\ w \equiv w_0 \pmod{|t|}\}, \qquad c_0^{\text{loc}} = tu_0 - sw_0. \tag{22}$$

Substituting the local level $c_0^{\text{loc}} + stk$ into (21) via x_0, y_0 translates it into the global level

$$(Qx_0 - Py_0) + n_1 n_2 \big(c_0^{\text{loc}} + stk\big) = c_0 + n_1 n_2 stk = c_0 + PQk, \tag{23}$$

where the first equality uses $c_0 = (Qx_0 - Py_0) + n_1 n_2 c_0^{\text{loc}}$ (a direct computation confirming this matches the formula for c_0 stated in (20)), and the second uses $PQ = n_1 n_2 st$. Combining (22) and (23) with (21) gives exactly (20). □

Lean formalization status: ***formally verified.*** *Theorem `LatticeLineCovers.splitting`. Matching the paper's hypotheses exactly required adding $0 < n_1$, $0 < n_2$, $s \neq 0$, $t \neq 0$ (needed for `mul_left_cancel₀`-style cancellation in the proof, and genuinely present, if implicitly, in the informal statement above via "coprime* positive *integers n_1, n_2" and "*nonzero *integers s, t") — this omission was caught and fixed* before *delegating the proof, not after. The Lean proof runs two parallel coprimality-transfer arguments (one per coordinate), the same technique as Lemma 8's proof. As with Lemma 7, $s \neq 0$ and $t \neq 0$ turned out to be unused by the actual Lean proof (only $n_1, n_2 \neq 0$ are load-bearing for the cancellations) — again confirmed via the unused-variable linter, and again kept in the statement for fidelity to this paper's own hypotheses rather than removed. Independently re-verified axiom-clean.*

Example 11. Take $n_1 = 2$, $n_2 = 3$, $x_0 = y_0 = 1$, so $R = R(2,3,1,1) = \{(x,y) : x \text{ odd},\ y \equiv 1 \,(\mathrm{mod}\, 3)\}$. Take $s = 2$, $t = 3$; one checks directly that (17) holds:

$$\gcd(s,n_2) = \gcd(2,3) = 1, \quad \gcd(t,n_1) = \gcd(3,2) = 1, \quad \gcd(s,t) = \gcd(2,3) = 1. \tag{24}$$

Then $P = n_1 s = 4$, $Q = n_2 t = 9$ (and indeed $\gcd(4,9) = 1$). Take the residue pair $(u_0,w_0) = (1,2)$; then

$$c_0 = Q(x_0 + n_1 u_0) - P(y_0 + n_2 w_0) = 9\cdot(1+2) - 4\cdot(1+6) = 27 - 28 = -1. \tag{25}$$

Three points of $R_{1,2}$, at three different values of k in (20):

$$\begin{aligned}
(u,w) = (1,2) &\longmapsto (x,y) = (3,7): & \varphi_{(4,9)}(3,7) &= 27 - 28 = -1 = c_0 + 36\cdot 0,\\
(u,w) = (3,2) &\longmapsto (x,y) = (7,7): & \varphi_{(4,9)}(7,7) &= 63 - 28 = 35 = c_0 + 36\cdot 1,\\
(u,w) = (1,5) &\longmapsto (x,y) = (3,16): & \varphi_{(4,9)}(3,16) &= 27 - 64 = -37 = c_0 + 36\cdot(-1),
\end{aligned} \tag{26}$$

using $PQ = 36$ throughout.

Lemma 12 (Splitting: partition). *With R, s, t as in Lemma 10,*

$$R = \bigsqcup_{\substack{0\le u_0<|s|\\ 0\le w_0<|t|}} R_{u_0,w_0} \tag{27}$$

(a disjoint union over the $|s||t|$ residue pairs).

Pedantic example (working notes only — not for the final article). (i) Conclusion holds. With $n_1 = 2, n_2 = 3, x_0 = y_0 = 1, s = 2, t = 3$ as in Example 11, R is partitioned into $|s||t| = 6$ pieces R_{u_0,w_0}, $0 \le u_0 < 2$, $0 \le w_0 < 3$. The point $(x,y) = (7,7)$ has internal coordinates $(u,w) = (3,2)$ (i.e. $7 = 1 + 2\cdot 3$, $7 = 1 + 3\cdot 2$), so its residue pair is $u_0 =$

$3 \bmod 2 = 1$, $w_0 = 2 \bmod 3 = 2$: it lies in $R_{1,2}$ and in *no other* of the 6 pieces, matching the "$\exists!$" (exists-uniquely) content of the lemma. **(ii) Why $s \neq 0$ cannot be dropped.** With $s = 0$ (and, say, $t = 3$ as before), the index range $0 \le u_0 < |s| = 0$ is *empty*, so the right-hand side of (27) is an empty union $\bigsqcup_\varnothing = \varnothing$ — yet R itself is nonempty (e.g. $(1,1) \in R$), so the claimed equality fails outright. (Concretely, $s = 0$ also makes R_{u_0,w_0} from (19) ill-posed, since "$u \equiv u_0 \pmod{|s|}$" is congruence modulo 0, i.e. $u = u_0$ exactly — a single internal-coordinate value, not a residue class — so the whole finite-partition statement, which relies on exactly $|s|$ residue classes covering all of $\mathbb{Z}$, has no content left to state.) This is exactly why s, t nonzero is carried over as a standing hypothesis from Lemma 10.

Proof. Fix $(x, y) \in R$, with internal coordinates (u, w) as in Definition 9. By the division algorithm, $u = s\lfloor u/s \rfloor + u_0$ for a unique u_0 with $0 \le u_0 < |s|$ (interpreting the division algorithm with respect to $|s|$; concretely, u_0 is the unique representative of u modulo s lying in $[0, |s|)$, and similarly $w_0 \in [0, |t|)$ for w modulo t). By construction $u \equiv u_0 \pmod{|s|}$ and $w \equiv w_0 \pmod{|t|}$, so $(x, y) \in R_{u_0,w_0}$ for this particular pair (u_0, w_0), establishing $R \subseteq \bigcup_{u_0,w_0} R_{u_0,w_0}$; the reverse inclusion is immediate from (19). For disjointness (equivalently, uniqueness of (u_0, w_0) for each point): suppose $(x, y) \in R_{u_0,w_0} \cap R_{u_{0'},w_{0'}}$ via internal coordinates (u, w) and (u'', w'') respectively. Since internal coordinates are unique (Definition 9), $u'' = u$ and $w'' = w$, so $u_0 \equiv u \equiv u_{0'} \pmod{|s|}$; as both $u_0, u_{0'} \in [0, |s|)$, an interval of length $|s|$, and their difference is a multiple of s of absolute value $< |s|$, we get $u_0 = u_{0'}$, and symmetrically $w_0 = w_{0'}$. □

*Lean formalization status: **formally verified.** Theorem `LatticeLineCovers.splitting_partition`, phrased as existence-and-uniqueness of the residue pair (u_0, w_0) for every point of R (an `ExistsUnique` statement, Lean/Mathlib notation $\exists!$), which is definitionally what a disjoint union means. This is pure Euclidean division together with the "two representatives of one residue class in the same length-$|m|$ window coincide" fact used in the last paragraph of the proof above; no coprimality is involved anywhere. A first, undelegated attempt at this lemma hit real friction — a cited lemma that does not exist in the current Mathlib version, and several automated (`omega`) tactic calls given goals it cannot discharge, since `omega` handles only linear integer arithmetic and the "length-$|m|$ window" argument genuinely needs one multiplicative step. Reverted rather than continuing to guess lemma names, and delegated (with the specific failures included in the brief so they were not repeated) to a fresh Opus-model subagent, which closed the multiplicative step via `abs_mul` together with `le_mul_of_one_le_right` and `Int.one_le_abs`, keeping `omega` strictly to the resulting linear facts. Independently re-verified axiom-clean.*

This is where the earlier, flawed version of this argument failed: it attempted to realize an *arbitrary* target direction by using a multiple of it known to lie in the current sublattice, without checking that the multiple's own geometric line (which follows its *primitive* direction, not the multiple used to construct it) stays confined to the intended coset. Lemma 10 avoids this by construction, via the primitivity argument in its own proof.

Lemma 13 (Steering). *Let n_1, n_2 be coprime positive integers. For every $\theta \in [0, \pi)$ and every $\varepsilon > 0$ there exist nonzero integers s, t, $|s|, |t| \ge 2$, satisfying (17), such that the direction of $(n_1 s, n_2 t)$ lies within ε of θ.*

Pedantic example (working notes only — not for the final article). (i) Conclusion holds, fully numeric. Take $n_1 = 2, n_2 = 3$ (coprime), target $\mu = \tan\theta = 1$, tolerance $\varepsilon = 0.1$. From (28): $\delta = \frac{1}{2}(1 - \frac{1}{2}) = \frac{1}{4}$, $E = 2^{\omega(2)+1} = 2^2 = 4$; from (30): $L_0 = \lceil (4+1)/\frac{1}{4} \rceil = 20$. Condition (31) needs $20 + 2 < 0.1 \cdot 2 \cdot p/3 = p/15$ (note: no μ in this condition — see the remark right after (31) in the proof below), i.e. $p > 330$; take the prime $p = 331$ ($331 \nmid n_1 n_2 = 6$), so $s = 331$. Then $t_0 = \lfloor 1 \cdot 2 \cdot 331/3 \rfloor + 2 = \lfloor 220.67 \rfloor + 2 = 222$, and searching $I = [222, 242)$ for t coprime to $n_1 p = 662$: $t = 223$ (prime) works ($\gcd(223, 2) = 1$, $\gcd(223, 331) = 1$), and also $\gcd(s, t) = \gcd(331, 223) = 1$, $\gcd(s, n_2) = \gcd(331, 3) = 1$, $\gcd(t, n_1) = \gcd(223, 2) = 1$, so (17) holds; note $t = 223 \ge 2$ automatically, by construction of the shifted window (32), with no separate check needed. The resulting slope is $n_2 t/(n_1 s) = 669/662 \approx 1.01057$, within $\varepsilon = 0.1$ of $\mu = 1$ as promised. **(ii) A subtlety about $\gcd(n_1, n_2) = 1$: not used inside *this* proof, but still essential downstream.** Repeat the same computation with the non-coprime pair $n_1 = 4, n_2 = 6$ ($\gcd = 2$) — note 4 has the same radical ($\{2\}$) as $n_1 = 2$ above, so δ, E, L_0 and even the threshold on p are literally unchanged, and $p = 331$, $t = 223$ still satisfy every step of the argument verbatim, giving slope $n_2 t/(n_1 s) = 1338/1324 \approx 1.01057$, again within ε of μ: (29)–(33) never once refer to $\gcd(n_1, n_2)$, so nothing in *this* proof breaks. What *does* break is downstream: $(P, Q) = (n_1 s, n_2 t) = (1324, 1338)$ has $\gcd(1324, 1338) = 2$ — not a primitive direction at all, so it could not legally be fed into Lemma 10 (which needs $\gcd(n_1, n_2) = 1$ among its own standing hypotheses to guarantee $\gcd(P, Q) = 1$) or used as a line direction under Definition 2. So $\gcd(n_1, n_2) = 1$ is carried by Lemma 13 as a hypothesis not because its own proof needs it, but because it is the invariant maintained by the surrounding recursive construction (Theorem 6's proof) that every lemma along the way is entitled to assume. **(iii) A real bug this pedantic pass actually found, now fixed.** An earlier version of (31) read $L_0 < \varepsilon\mu n_1 p/n_2$ (an extra, wrong factor of μ). Concretely, with $n_1 = n_2 = 1$, $\mu = 100$, $\varepsilon = 0.01$: that wrong condition gives $\delta = \frac{1}{2}$, $E = 2$, $L_0 = 6$, threshold $p > 6$, so $p = 7$; the best t in the resulting window ($t_0 = 700$, $I = [700, 706)$) is $t = 701$, giving slope $701/7 \approx 100.142857$ — off from $\mu = 100$ by ≈ 0.143, over $14\times$ the claimed tolerance $\varepsilon = 0.01$, not a rounding-level near-miss. The wrong condition is even *unsatisfiable* at $\mu = 0$ (reduces to $L_0 < 0$). Both failures trace to the same algebra error: $L_0 < \varepsilon\mu n_1 p/n_2$ only implies $(n_2 L_0)/(n_1 p) < \varepsilon\mu$, not $< \varepsilon$, and these coincide only when $\mu \le 1$. The corrected, μ-free condition (31) used in the proof below fixes this: with the same $n_1, n_2, \mu, \varepsilon$, it gives threshold $p > 800$, so $p = 809$; the best t in $I = [80902, 80908)$ is $t = 80902$, giving slope $80902/809 \approx 100.002472$, comfortably within $\varepsilon = 0.01$. (Script: `code/steering_bug_check.py`; log: `runs/2026-08-02_steering_bug_check.md`.) **(iv) A second, independent bug found on the next adversarial pass, now fixed: window frozen at $\mu = 0$.** Reuse $n_1 = 2, n_2 = 3$ so $\delta = \frac{1}{4}$, $E = 4$, $L_0 = 20$ as in (i). At $\mu = 0$, the *unshifted* window used by an earlier version of this proof was $t_0 = \lfloor 0 \cdot n_1 p/n_2 \rfloor = 0$ for every prime p, i.e. $I = [0, 20)$ regardless of p: taking $p = 307$ gives $I = [0, 20)$, and so does the much larger $p = 100003$ — unlike the $\mu = 1$ case in (i), where a larger p pushes t_0 further from the origin, at $\mu = 0$ the window never moves. Since $\gcd(1, n_1 p) = 1$ always, $t = 1 \in I$ is always a valid witness of "≥ 1 coprime integer in I," and is in fact the *closest-to-target* candidate (it minimizes $|n_2 t/(n_1 p) - 0|$ among positive t) — yet $t = 1$ violates the lemma's own $|t| \geq 2$ requirement; the earlier proof's "(for p large) $|t| \geq 2$" was simply asserted, not derived, and at $\mu = 0$ no amount of growing p makes it true on its own. (For this particular $n_1 = 2$, other coprime witnesses with $|t| \geq 2$ do also exist in $[0, 20)$, e.g. $t = 3$ or $t = 5$, so the *lemma* was never false here — only the proof's justification for picking a witness was incomplete.) The shifted window (32), $t_0 = \lfloor \mu n_1 p/n_2 \rfloor + 2$, fixes this structurally: at $\mu = 0$ it gives $t_0 = 2$ for every p, so $I = [2, 22)$ contains only integers with $|t| \geq 2$ from the outset, and the same sieve count $\delta L_0 - E \geq 1$ now guarantees a genuinely usable witness with no largeness argument needed. (Script: `code/steering_mu0_bug_check.py`; log: `runs/2026-08-02_steering_mu0_bug_check.md`.)

Proof. By symmetry (swap the roles of s and t to treat $\theta = \pi/2$, which targets reciprocal slope 0 and is handled by the same argument below with the roles of n_1, n_2 exchanged, subject to the same window shift introduced at (32)) it suffices to treat $\theta \neq \pi/2$, i.e. a finite target slope $\mu = \tan\theta$; the case $\mu < 0$ is identical with $t \mapsto -t$, so assume $\mu \geq 0$.

Set

$$\delta = \frac{1}{2} \prod_{r \mid n_1} \left(1 - \frac{1}{r}\right) > 0, \quad E = 2^{\omega(n_1)+1}, \tag{28}$$

both depending only on n_1, where $\omega(n_1)$ is the number of distinct prime divisors of n_1. For a prime $p \nmid n_1 n_2$ and any interval $I \subset \mathbb{Z}$ of length L, inclusion–exclusion over the divisors of $n_1 p$ gives

$$\#\{t \in I : \gcd(t, n_1 p) = 1\} \geq L \prod_{r \mid n_1} \left(1 - \frac{1}{r}\right)\left(1 - \frac{1}{p}\right) - 2^{\omega(n_1)+1} \geq \delta L - E, \tag{29}$$

using $p \geq 2$ for the last factor. Fix

$$L_0 = \left\lceil \frac{E+1}{\delta} \right\rceil, \tag{30}$$

independent of p — and, crucially, independent of μ as well. Choose a prime $p \nmid n_1 n_2$ large enough that

$$L_0 + 2 < \frac{\varepsilon n_1 p}{n_2} \tag{31}$$

(possible since the right side $\to \infty$ as $p \to \infty$, for fixed ε, n_1, n_2; note (31) does not involve μ at all, unlike t_0 below), set $s = p$ and

$$t_0 = \left\lfloor \frac{\mu n_1 p}{n_2} \right\rfloor + 2. \tag{32}$$

The $+2$ shift in (32), absent from an earlier version of this proof, is what makes the forthcoming $|t| \geq 2$ requirement hold *unconditionally* rather than only "for p large": since $\mu \geq 0$, already $t_0 \geq 2$ by (32), so every integer in the window I below inherits $|t| \geq 2$ directly, with no degenerate behaviour at $\mu = 0$ (where, without the shift, $t_0 = 0$ for *every* p, the window never moves away from the origin as $p \to \infty$, and $t = 1$ — always coprime to $n_1 p$ — is a witness that nothing rules out being the *only* one, violating $|t| \geq 2$). The interval $I = [t_0, t_0 + L_0)$ then contains, by (29) with $L = L_0$ and (30), at least $\delta L_0 - E \geq 1$ integer t with $\gcd(t, n_1 p) = 1$ — equivalently $\gcd(t, n_1) = \gcd(t, s) = 1$, as $s = p$ is prime — and, as just noted, $|t| \geq 2$ automatically. For any such t,

$$\left| \frac{n_2 t}{n_1 s} - \mu \right| = \frac{n_2}{n_1 p} \left| t - \frac{\mu n_1 p}{n_2} \right| < \frac{n_2 (L_0 + 2)}{n_1 p} < \varepsilon, \tag{33}$$

using (31) for the last step (and $|t - \mu n_1 p/n_2| < L_0 + 2$, which follows from $t \in [t_0, t_0 + L_0)$ and $t_0 - \mu n_1 p/n_2 \in (1, 2]$ by (32)), so $(P, Q) = (n_1 s, n_2 t)$ has slope within ε of μ, hence (as arctan is 1-Lipschitz, and $\arctan\mu = \theta - \pi$ when $\theta > \pi/2$ identifies the same point of $\mathbb{RP}^1$) direction within ε of θ. □

Lean formalization status: ***not yet formalized.*** *Flagged, before any Lean work on this project began, as the single hardest piece of the whole formalization: it is a genuine (if elementary) analytic-number-theory argument, not pure algebra like Lemmas 7–12. Worth checking, before attempting to reprove (29) from scratch in Lean, whether Mathlib's existing sieve-theory development (e.g.* `Mathlib.NumberTheory.SelbergSieve` *or similar) already covers a bound of this shape.*

The point of Lemma 13 is that it is *not* enough to know that *some* admissible (s, t) exists at every stage: repeatedly reusing, say, $(s, t) = (2, 3)$ is admissible forever (all three coprimality conditions of (17) hold trivially against the resulting $n_1 = 2^k, n_2 = 3^k$), yet gives directions converging to a single point of $\mathbb{RP}^1$, not a dense set. What is needed, and what Lemma 13 provides, is the ability to steer toward an *arbitrary prescribed* target at every stage.

Remark 14 (Tails of a dense sequence remain dense). If X is a T_1 topological space with no isolated points, $D \subseteq X$ is dense, and $F \subset X$ is finite, then $D \setminus F$ is still dense in X. (Any nonempty open U is itself infinite, since no point of X is isolated; so $U \setminus F$ is nonempty and open, hence meets D, giving a point of $D \setminus F$ in U.) Applied with $X = \mathbb{RP}^1$ (compact, metric, hence T_1, and with no isolated points as a circle) and $D = \{\theta_k\}_{k \geq 1}$: every tail $\{\theta_k\}_{k \geq N}$ is dense, since it contains $D \setminus$

$\{\theta_1, \ldots, \theta_{N-1}\}$. This is the fact the density argument below needs beyond mere density of $\{\theta_k\}$ itself — without "no isolated points" it can fail (e.g. a sequence dense in $\{0\} \cup [1, 2]$ that visits 0 only once).

Proof of Theorem 6. Fix an enumeration $(z_k)_{k\geq 1}$ of $\mathbb{Z}^2$ and a sequence $(\theta_k)_{k\geq 1}$ dense in $[0, \pi)$ (e.g. $\theta_k = k\alpha \bmod \pi$ for α/π irrational, by equidistribution of the irrational rotation). Set $R_0 = \mathbb{Z}^2$, i.e. $(n_1, n_2, x_0, y_0) = (1, 1, 0, 0)$.

Given $R_{k-1} = R(n_1, n_2, x_0, y_0)$, apply Lemma 13 to find s, t (possibly negative — Lemma 13 allows this, needed to reach both signs of slope) realizing a direction within $1/k$ of θ_k, and apply Lemmas 10–12 to split R_{k-1} into its $|s||t| \geq 4$ sub-cosets, each fully coverable by direction $(P_k, Q_k) = (n_1 s, n_2 t)$ restricted to one residue class of levels. Put into $\mathcal{F}$ all such lines for every sub-coset except one, reserved as follows: if $z_k \in R_{k-1}$, reserve the lexicographically-least (u_0, w_0) whose sub-coset does not contain z_k (possible since there are ≥ 4 sub-cosets and only one need be avoided); otherwise reserve the lexicographically-least (u_0, w_0) outright — a concrete rule is needed for the recursion to be a well-defined function, as is a concrete choice among the (s, t) that Lemma 13 only asserts to exist — both are choice functions to be supplied explicitly in a Lean formalization, not just existence claims. Let $R_k = R_{u_0,w_0}$ be that reserved sub-coset. The two quantities that matter going forward are genuinely different: the *achieved direction* recorded for the density argument below is the (possibly signed) pair $(P_k, Q_k) = (n_1 s, n_2 t)$, while R_k itself, to serve as the input $R(n_{1'}, n_{2'}, x_{0'}, y_{0'})$ to the *next* stage's application of Lemma 13 (whose hypotheses require positive $n_{1'}, n_{2'}$, as does Definition 9), must use the *positive* pair $n_{1'} = n_1|s|$, $n_{2'} = n_2|t|$, $x_{0'} = x_0 + n_1 u_0$, $y_{0'} = y_0 + n_2 w_0$: a direct check from the definition of R_{u_0,w_0} (19) (writing $u = u_0 + |s|u'$, $w = w_0 + |t|w'$ for the internal coordinates, since "$u \equiv u_0 \pmod{|s|}$" means exactly this) shows $R_{u_0,w_0} = R(n_1|s|, n_2|t|, x_{0'}, y_{0'})$ on the nose. Write $R_k = R\left(n_1^{(k)}, n_2^{(k)}, x_0^{(k)}, y_0^{(k)}\right)$ for this representation, i.e. $n_1^{(k)} = n_1|s|$, $n_2^{(k)} = n_2|t|$ (with $n_1^{(0)} = n_2^{(0)} = 1$ from the base case), the notation used below. This keeps $\gcd\left(n_1^{(k)}, n_2^{(k)}\right) = \gcd(n_1|s|, n_2|t|) = \gcd(P_k, Q_k) = 1$ (Lemma 10), so the invariant needed for the next iteration is preserved.

Pedantic example (working notes only — not for the final article). Another real gap this pedantic pass found, now fixed above. An earlier version of this proof used $(P_k, Q_k) = (n_1 s, n_2 t)$ directly, unmodified, as the next stage's (n_1, n_2) pair — silent, because it never mattered until s or t was actually negative. Concretely: start from $R_0 = R(1, 1, 0, 0) = \mathbb{Z}^2$, and suppose stage 1 picks $s = -3, t = 2$ (both satisfy $|s|, |t| \geq 2$; $\gcd(s, n_2) = \gcd(-3, 1) = 1$, $\gcd(t, n_1) = \gcd(2, 1) = 1$, $\gcd(s, t) = \gcd(-3, 2) = 1$, so (17) holds), giving $(P_1, Q_1) = (-3, 2)$ — a perfectly good, primitive, achieved direction ($\gcd(-3, 2) = 1$) to record for density purposes. But Definition 9 and Lemma 13 both require their n_1, n_2 *positive*; feeding $n_{1'} = -3$ into either is simply outside what they're stated for, not merely inconvenient. The fix used above — $n_{1'} = |-3| = 3$, $n_{2'} = |2| = 2$ — is not a different choice, just the *correct name* for the same set: taking $(u_0, w_0) = (0, 0)$ for concreteness, $R_{0,0} = \{(x, y) : x \equiv 0 \pmod 3,\ \ y \equiv 0 \pmod 2\} = R(3, 2, 0, 0)$, exactly matching $(n_1|s|, n_2|t|) = (3, 2)$ and manifestly *not* matching $(P_1, Q_1) = (-3, 2)$ read literally as a positive pair.

Coverage invariant. Stage i claims exactly $R_{i-1} \setminus R_i$, and $R_i \subseteq R_{i-1}$ for every $i \geq 1$. Telescoping from $R_0 = \mathbb{Z}^2$ gives, for every k,

$$\mathbb{Z}^2 \setminus R_{k-1} = \bigcup_{i<k} (R_{i-1} \setminus R_i) = \bigcup_{i<k} (\text{stage-}\ i\ \ \text{claims}), \tag{34}$$

i.e. a point lies outside the current region R_{k-1} exactly when some earlier stage already claimed it. This is what justifies the reservation rule above only ever needing to avoid z_k when $z_k \in R_{k-1}$: whenever $z_k \notin R_{k-1}$, (34) shows z_k was already claimed by an earlier stage, so no action is needed for it at stage k.

Pedantic example (working notes only — not for the final article). Why the coverage argument needs the invariant (34), fully numeric. Take $z_5 = (0, 0) \in \mathbb{Z}^2$, and suppose (hypothetically) $R_4 = R(4, 3, 1, 0)$, so $z_5 = (0, 0) \notin R_4$ (since $0 \neg\equiv 1 \pmod 4$). The reservation rule at stage 5 only ever looks at membership in R_4 — since $z_5 \notin R_4$, stage 5 does nothing to claim z_5 specifically. For the overall coverage claim $\bigcup_k (\text{stage-}\ k\ \ \text{claims}) = \mathbb{Z}^2$ to hold, z_5 must therefore already have been claimed at some *earlier* stage; (34) is exactly what proves this: $z_5 \notin R_4$ forces $z_5 \in R_{i-1} \setminus R_i$ for some $i \leq 4$. Without the invariant, the Coverage paragraph's "if $z_k \notin R_{k-1}$, it was already claimed at some earlier stage" step would be an unjustified assertion rather than a proven fact.

Coverage. By (34), every z_k is eventually claimed: if $z_k \in R_{k-1}$, the reservation rule ensures z_k lies in one of the claimed sub-cosets at stage k (not the reserved R_k); if $z_k \notin R_{k-1}$, it was already claimed at some earlier stage. So

$$\bigcup_k (\text{stage-}\ k\ \ \text{claims}) = \mathbb{Z}^2. \tag{35}$$

Disjointness. Every individual line placed into $\mathcal{F}$ at stage k has all of its lattice points inside the sub-coset R_{u_0,w_0} it came from (a subset of $R_{k-1} \setminus R_k$, since $R_{u_0,w_0} \neq R_k$): this is exactly (20) of Lemma 10 (each line on the left is a subset of the union), the step the earlier, flawed version of this construction got wrong (see the remark following Lemma 10). So stage k's claimed points lie in $R_{k-1} \setminus R_k \subset R_{k-1}$. For $j > k$, $R_{j-1} \subseteq R_k$ (as $R_i \subseteq R_{i-1}$ for every i), so stage j's points, lying in R_{j-1}, are disjoint from stage k's. Hence distinct lines from different stages never share a lattice point, and distinct lines of the same direction from the same stage are disjoint level sets of the same $\varphi_{(P_k,Q_k)}$, so never share one either. (Different stages in fact always realize different directions — $|P_k| = n_1^{(k)}$ is strictly increasing in k, since $n_1^{(k)} = n_1^{(k-1)}|s_k| \geq 2n_1^{(k-1)}$ — though the argument above does not need this.)

Density. Fix $\theta \in \mathbb{RP}^1$ and $\varepsilon > 0$. By 14, every tail $\{\theta_k\}_{k \geq N}$ is dense in $\mathbb{RP}^1$; so for any N there is some $k \geq N$ with θ_k within $\varepsilon/2$ of θ. Taking $N > 2/\varepsilon$, the corresponding achieved direction (P_k, Q_k) — within $1/k < \varepsilon/2$ of θ_k — then lies within ε of θ. (Each (P_k, Q_k) is genuinely realized in $\mathcal{F}$: at least $|s||t| - 1 \geq 3$ of the ≥ 4 sub-cosets at stage k are claimed, all nonempty, each contributing lines of that direction.) So every $\theta \in \mathbb{RP}^1$ is a limit of directions realized in $\mathcal{F}$: the realized direction set is dense.

Pedantic example (working notes only — not for the final article). Why the density argument needs a second use of density, fully numeric. A closure inference of the shape "$d(a_k, b_k) \leq 1/k$ for all k, so $\overline{\{a_k\}} \supseteq \overline{\{b_k\}}$" is false in general: take $b_1 = 0$, $b_k = 1$ for $k \geq 2$, and $a_k = 1$ for all k (legal, since $d(a_1, b_1) = 1 \leq 1/1$). Then $\overline{\{b_k\}} = \{0, 1\}$ but $\overline{\{a_k\}} = \{1\} \not\ni 0$: the value 0 is only ever shadowed at the one index where the tolerance $1/k$ is too coarse to matter. The fix used above blocks exactly this: instead of using *every* index k once, it selects, for each target θ and tolerance ε, an index k chosen *after* ε is fixed and large enough that both θ_k is already close to θ (using 14, which needs $\mathbb{RP}^1$ to have no isolated points) and $1/k$ is small. Applied to the toy example: there is no tail of $\{\theta_k\}$ avoiding the value $\theta = 0$ forever, so this style of argument could never reach the toy example's false conclusion.

□

Lean formalization status: ***not yet formalized****; see the status note under Theorem 6 above. Two write-up gaps found and fixed by adversarial review, 2026-08-02 (not affecting the truth of the theorem, only the rigor of this proof): the coverage argument originally left the case $z_k \notin R_{k-1}$ undischarged (fixed by (34)); the density argument's closure inference was a non-sequitur as originally stated (fixed by the explicit arbitrarily-large-k argument above). Before formalizing: the reservation and the Steering witness are existential choices, not yet functions — will need `Classical.choice`-style witnesses or an explicit deterministic rule (e.g. lexicographically-least valid choice) for the recursion to type-check as a definition.*

4. Formalization status: summary

Statement above	Lean theorem	Status
Lemma 7 (Rigidity)	`LatticeLineCovers.rigidity`	proved
Lemma 8 (Coset)	`LatticeLineCovers.coset`	proved
Lemma 10 (Splitting, coset identity)	`LatticeLineCovers.splitting`	proved
Lemma 12 (Splitting, partition)	`LatticeLineCovers.splitting_partition`	proved
Lemma 13 (Steering)	—	not started
Theorem 6 (main theorem)	—	not started

All four proved theorems are independently checked, via `#print axioms`, to depend only on Lean's three standard axioms (`propext`, `Classical.choice`, `Quot.sound`) — i.e. none of them rest on an unproved `sorry` anywhere in their dependency chain. The Lean source lives at `lean/LatticeLineCoversLean/Basic.lean` in this same repository; see `lean/STRATEGY.md` and `lean/PROOF.md` for the formalization strategy and a non-expert-oriented explanation of what was proved.

5. A picture

Figure 1 plots the first 10, 20, and 100 directions realized by an actual run of the construction (code in `code/`, targeting the golden-angle equidistributed sequence $\theta_k = k\pi\left(\sqrt{5} - 1\right) \bmod \pi$), each direction shown as an antipodal pair of points on the unit circle (a line's direction is unsigned) and colored by its position in the sequence.

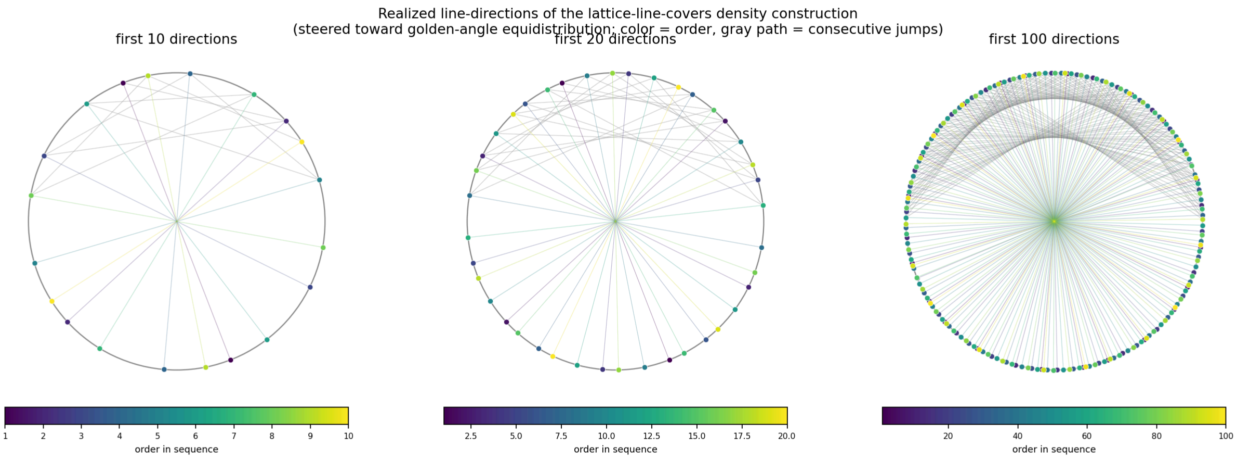

Figure 1: Realized directions after 10, 20, and 100 steps of the construction, colored by order (dark = early, light = late), with a thin path tracing consecutive jumps. The steering argument scatters rather than sweeps around the circle, since each target angle is chosen from an equidistributed sequence rather than visited in angular order.

Remark 15 (Verification, informal). Both the splitting mechanism (Lemmas 10–12) and the sign-generality of Lemma 13 were checked by direct brute-force simulation (exhaustive crossing and coverage checks on finite windows up to 81×81, including adversarial diagonal and sign-alternating schedules) before being trusted; an earlier version of the splitting step, without the primitivity safeguard built into Lemma 10, was found this way to be genuinely flawed rather than merely under-justified. The full argument was also read independently by a second reader working from the statement of the lemmas alone, who found a real gap in an earlier version of the density argument (restricting s, t to be positive only reaches half of $\mathbb{RP}^1$, and existence of *some* admissible move at each stage does not imply the ability to steer toward an *arbitrary* target — both addressed in the versions of Lemmas 10–12 and 13 given here). See the project logbook for the full history. This remark records the note's *informal* verification history, predating and independent of the formal Lean verification recorded in the status notes above; the two are complementary, not redundant — brute-force simulation and independent reading catch different classes of error than a machine-checked proof does, and vice versa.

6. Related work

The reformulation via φ_d and the resulting rigidity phenomenon (Lemma 7) is the natural two-dimensional analogue of the classical theory of *covering systems of congruences*, introduced by Erdős [3] to show that a positive proportion of the integers are not of the form $2^k + p$ for p prime. A very close relative of our covering problem — covering $\mathbb{Z}^2$ by finitely many finite-index *sublattices* (necessarily containing the origin, unlike the general affine cosets considered here) — is studied by Cremona and Koymans [1], who give it the same "projective/homogeneous covering congruences" framing; their refinement machinery (splitting one lattice into several of prime-power relative index) is structurally reminiscent of the splitting used here, one rank up.

A different classical result addresses a related but distinct question about how many directions a covering needs. Corzatt [2] conjectured that if a *finite, convex* set of lattice points is covered by n lines (with no constraint on where the lines may cross), the lines can always be chosen to use at most four distinct slopes. Our setting is the opposite regime — an infinite, unbounded point set, with a crossing-avoidance constraint standing in for a bound on the number of lines — and correspondingly the answer is qualitatively different: densely many directions are both necessary to consider and available.

The forbidden unimodular pairs of Lemma 7 are exactly the Farey-neighbor pairs of slopes; see e.g. Hardy and Wright [4, Ch. III] for the classical theory of Farey sequences and the closely related Stern–Brocot tree. This is precisely the toolkit used in the digital-geometry literature on digitized straight lines, where continued-fraction and Farey-tree techniques describe digital approximations to a line of given (rational or irrational) slope; see Kiselman [5], and, for the continued-fraction approach specifically, Uscka-Wehlou's dissertation [6], written under Kiselman's supervision at Uppsala University.

Open problem 16. Generalize Theorem 6 to $\mathbb{Z}^n$, $n \geq 3$: cover $\mathbb{Z}^n$ by hyperplanes (or lines, or general k-flats) such that no two of different direction meet at a lattice point, and determine which sets of directions can occur.

Acknowledgements

The results in this note — the reformulation, the rigidity and coset lemmas, the recursive construction, and both rounds of error-correction (one caught by numerical simulation, one by an independent review) — were found by Claude (Anthropic), working with Jan Snellman in an interactive session, in the course of an idle conversation about covering the lattice by lines. The additional references collected in the extended-references version of this note were located and verified by Claude via web search. This more formal, Lean-status-annotated version was likewise prepared by Claude, at Jan Snellman's request, once part of the argument had been formally verified.

References

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